Course Code: CSE422

Course Title: Artificial Intelligence

Project Title: Diabetes Prediction using different ML Models

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Introduction:

This is a Machine Learning Project which focuses on Making a Classification Model. The Model will classify a person's diabetic status on the basis of some features.

Dataset Description:

Source : Kaggle

https://www.kaggle.com/gargmanas/pima-indians-diabetes

Data Description

- It has 8 features and based on that the model will classify a person if s/he has diabetes or not.
- It's a classification problem because 2 distinct outcomes are coming.
- There are a total of 768 data points. All the features are quantitative.
- The dataset has no balancing problem. It is balanced.
- To check if correlation exists , we plotted the correlation matrix below. But we didn't notice much correlation. Hence, we could not drop the correlated column as it didn't exist.

  The below matrix is visualized by the Python provided Seaborn library.

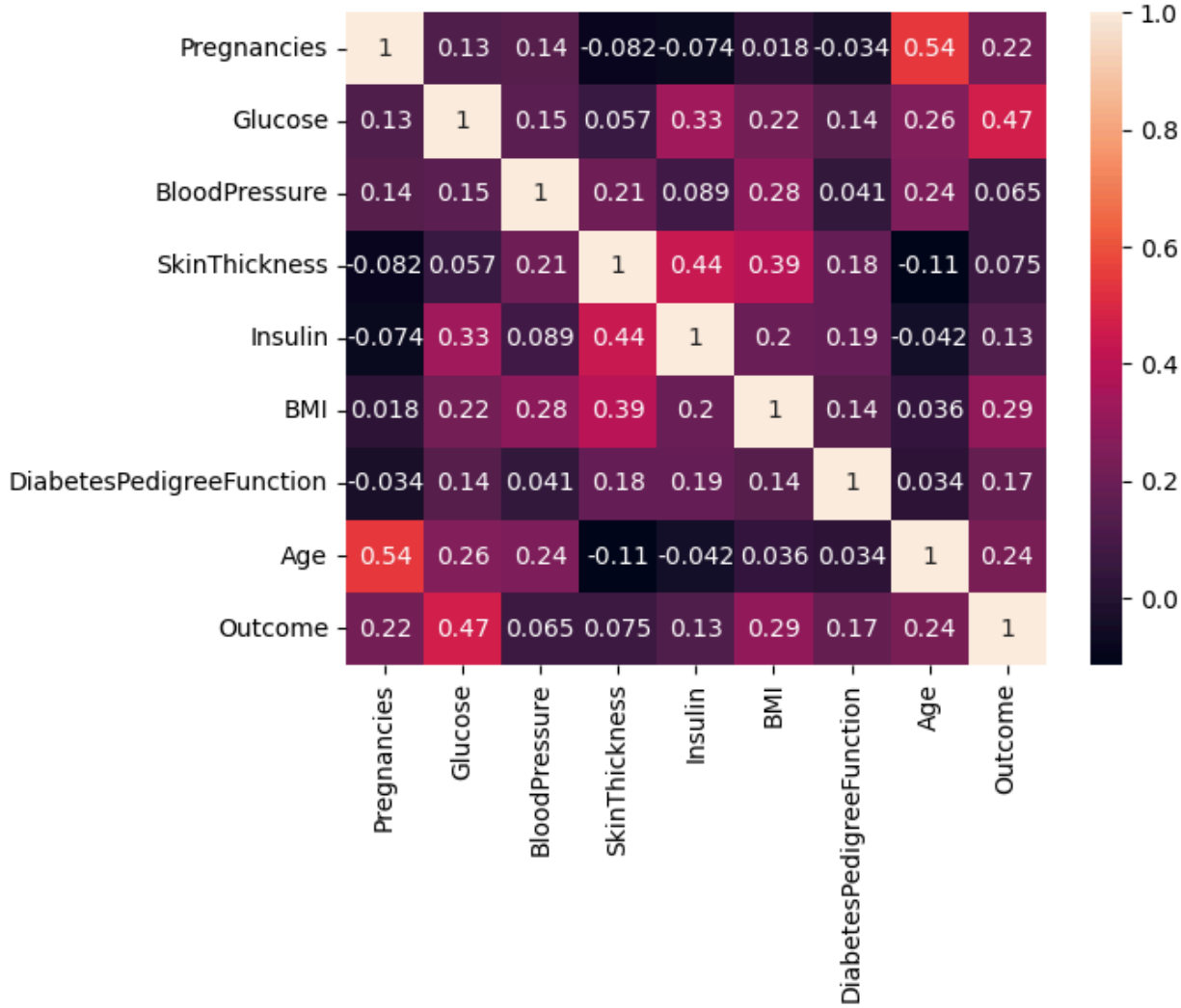

Figure: Correlation Matrix

Dataset Pre-Processing:

Faults:

The dataset has null values.

Solutions:

We filled the null values with the mean of the respective columns (Impute) as it is one of the techniques.

Feature Scaling:

The dataset had many negative values and values were arranged in different ranges. So, we used MinMax scaling to bring them in a specific range bound.

Dataset Splitting:

We used random state, training percentage as instructed in the template. That is, we took random states as 10 and train sets as 70%.

Model training and testing:

We used KNN, Decision Tree and Naive Bayes Classifier for model training and testing.
Among them Naive Bayes Classifier brought high accuracy with good precision and f1-score.

Model Selection/Comparison Analysis:

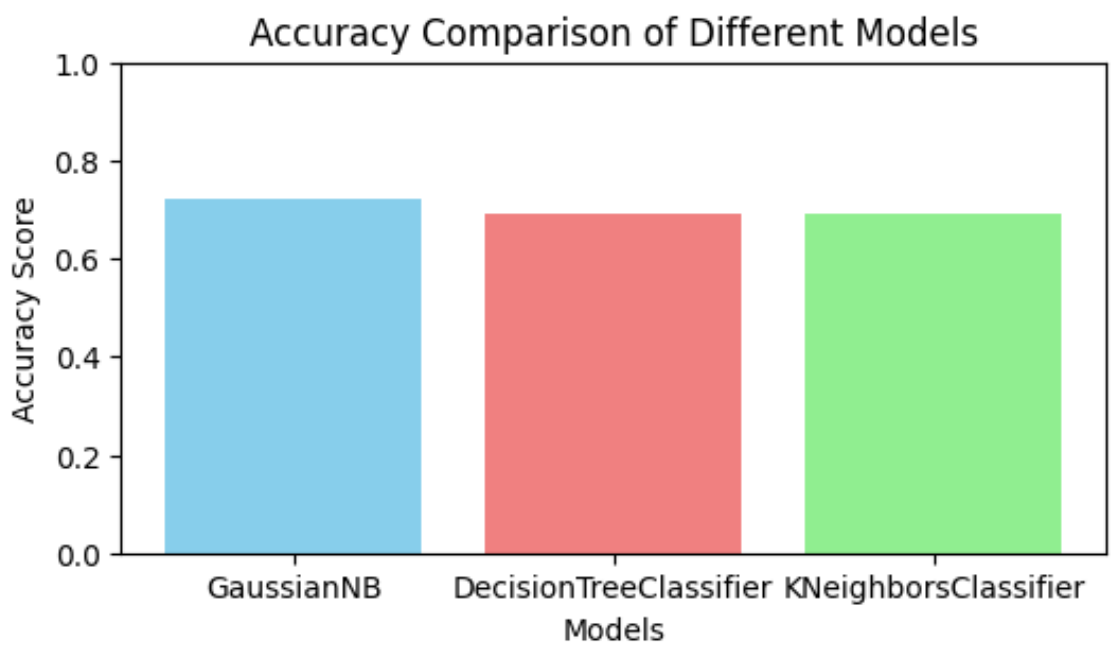

Figure: Accuracy Comparison of Different Models

Confusion Matrix:

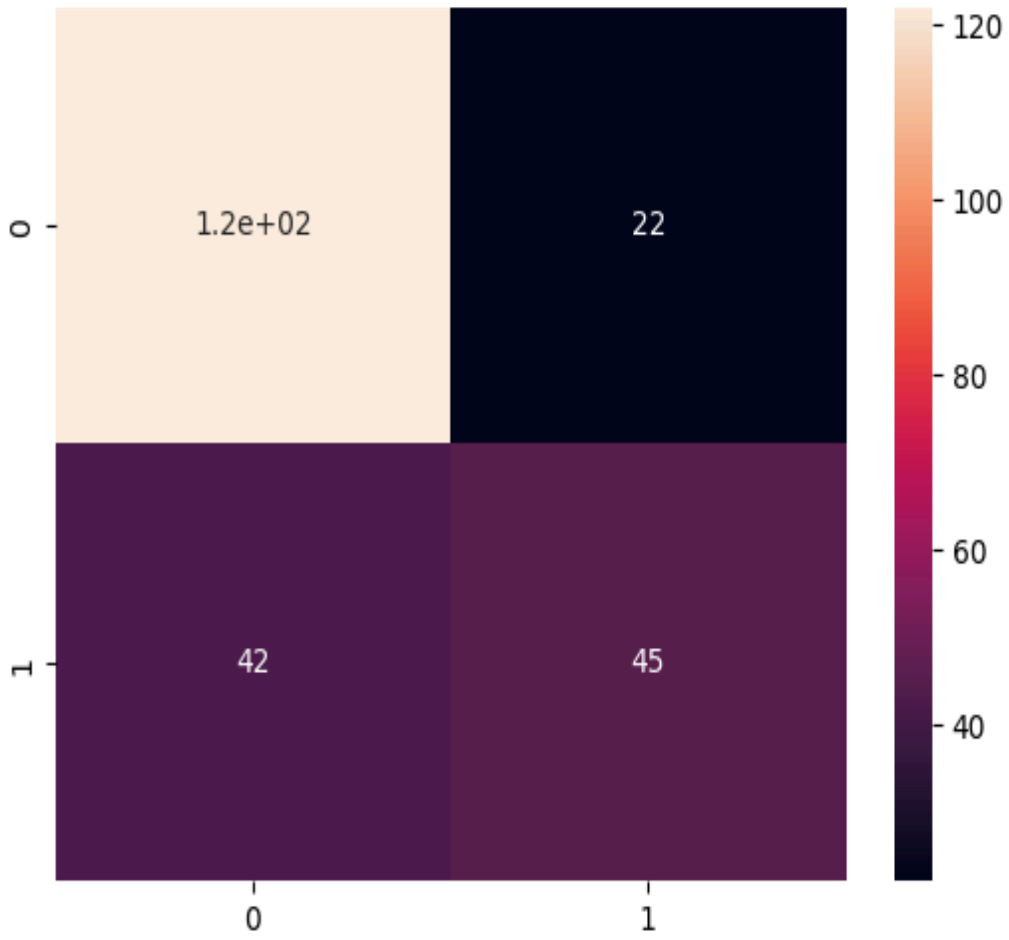

Figure: Confusion Matrix for Naive Bayes

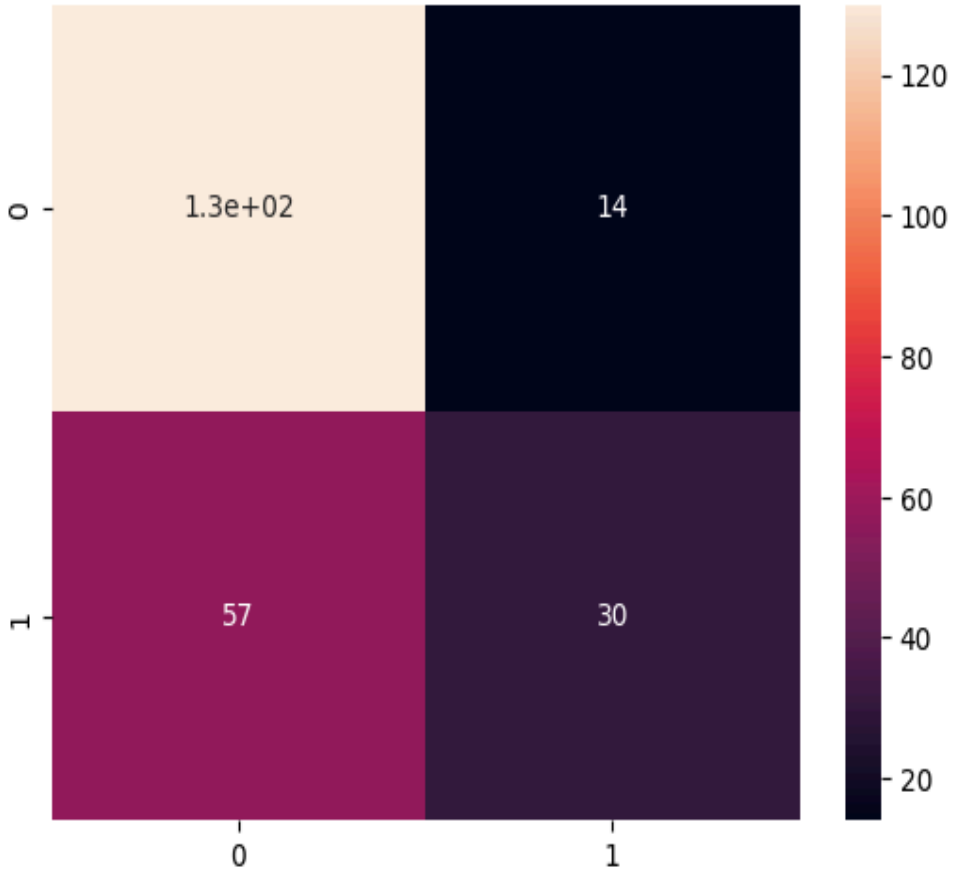

Figure: Confusion Matrix for KNN Classifier

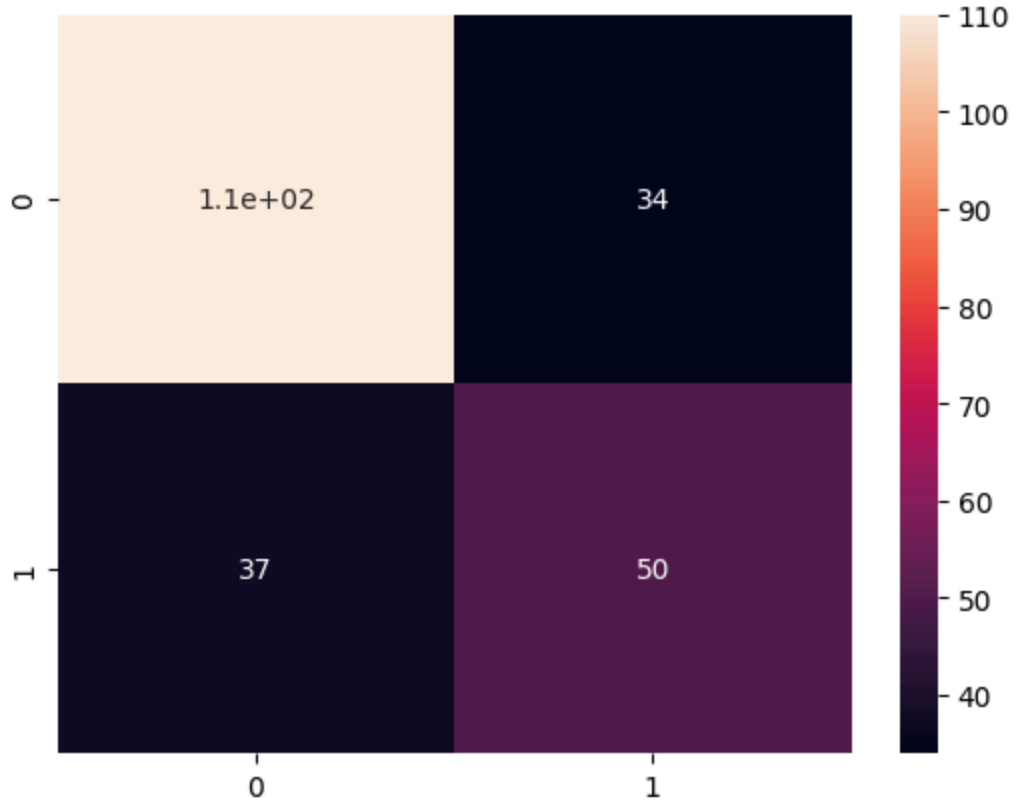

Figure: Confusion Matrix for DecisionTree Classifier

Precision, recall comparison of each model:

	precision	recall	f1-score	support
0	0.74	0.85	0.79	144
1	0.67	0.52	0.58	87
accuracy			0.72	231
macro avg	0.71	0.68	0.69	231
weighted avg	0.72	0.72	0.71	231

Figure: Precision, recall comparison of Naive Bayes

	precision	recall	f1-score	support
0	0.70	0.90	0.79	144
1	0.68	0.34	0.46	87
accuracy			0.69	231
macro avg	0.69	0.62	0.62	231
weighted avg	0.69	0.69	0.66	231

Figure: Precision, recall comparison of KNN Classifier

	precision	recall	f1-score	support
0	0.75	0.76	0.76	144
1	0.60	0.57	0.58	87
accuracy			0.69	231
macro avg	0.67	0.67	0.67	231
weighted avg	0.69	0.69	0.69	231

Figure: Precision, recall comparison of Decision Tree

Conclusion:

To conclude, we suggest that this type of project should regularly be updated keeping pace with the change in the features used to predict a person's diabetic status. And, while working for the project, we really had fun and we gathered experience on how Machine Learning actually works. Because the project required us to go through each step of a ML model development system. We look forward to working more on such projects in the days to come.